AI YouTube Learning Assistant

PROJECT REPORT

AIML Internship — 2026

Student	Numair Fahad
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Tech Stack	FastAPI · React · Google Gemini · ChromaDB · Clerk Auth
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1. Introduction

LearnTube is a full-stack AI-powered web application that transforms any YouTube video into an interactive, structured learning experience. Users paste a YouTube URL and instantly receive an AI-generated summary, chapter breakdowns, a multiple-choice quiz, flip-card flashcards, and a context-aware chat assistant — all grounded in the actual video transcript.

The project was built as the capstone of the AIML Internship program, demonstrating practical application of Retrieval-Augmented Generation (RAG), large language models, vector databases, and modern full-stack engineering practices.

Core capabilities at a glance:

■ **RAG Chat**	Ask any question about the video; answers are grounded in transcript excerpts with timestamp citations.
■ **Smart Summary**	Auto-generated overview, key points, and inferred chapter structure.
■ **AI Quiz**	Adaptive multiple-choice questions with explanations to test comprehension.
■ **Flashcards**	Spaced-repetition-ready flip cards covering key concepts and terms.
■ **User Auth**	Secure Clerk authentication with Google OAuth and email/password.
■ **Video Library**	Persistent per-user library to revisit and re-learn from any added video.

2. Problem Statement

YouTube hosts over 800 million videos and serves more than 1 billion hours of content daily. Despite being the world's largest free educational platform, learners face significant barriers to effective knowledge retention:

- Passive consumption — viewers watch without active recall or structured review.
- No quick-reference summaries — rewatching an entire video to recall a single point is inefficient.
- Lack of comprehension verification — no quizzes, exercises, or knowledge checks.
- No interactive Q&A; — learners cannot ask follow-up questions about specific content.
- Time inefficiency — long videos bury key concepts without navigation aids.
- Fragmented note-taking — learners must context-switch between video and note tools.

Our Solution

LearnTube solves all six barriers in a single unified interface. By extracting the transcript of any YouTube video, chunking it semantically, embedding it into a vector store, and feeding relevant chunks to a Google Gemini LLM, the application provides instant, grounded, interactive learning tools — without requiring any manual note-taking or video replay.

3. System Architecture

LearnTube follows a clean three-layer architecture: a React/Vite single-page application on the frontend, a FastAPI REST backend, and a persistence layer combining SQLite (relational data) with ChromaDB (vector embeddings).

Architecture Overview

Layer	Technology	Responsibility
Frontend	React 18 + Vite + Tailwind CSS v4	SPA UI, routing, state management
Authentication	Clerk (Replit-managed)	JWT sessions, Google OAuth, email/password
API Gateway	FastAPI (Python 3.13)	REST endpoints, auth middleware, RAG orchestration
LLM	Google Gemini 2.0 Flash	Chat answers, summary, quiz, flashcard generation
Vector Store	ChromaDB	Semantic chunk storage and similarity search
Relational DB	SQLite + SQLAlchemy	User library, video metadata, summaries
Transcript API	youtube-transcript-api	Fetches auto/manual captions from YouTube
Embeddings	sentence-transformers	all-MiniLM-L6-v2 local embedding model
Intent Classifier	scikit-learn TF-IDF + LR	Routes user queries to correct pipeline

RAG Pipeline Flow

When a user submits a YouTube URL, the backend executes the following pipeline:

Step 1 — Transcript Fetch	The youtube-transcript-api retrieves timestamped captions (auto or manual).

Step 2 — Chunking	Transcript is split into ~200-word semantic chunks, each tagged with a start timestamp.
Step 3 — Embedding	each chunk is embedded via sentence-transformers (all-MiniLM-L6-v2) and stored in ChromaDB.
Step 4 — RAG Query	User questions trigger a top-5 similarity search; retrieved chunks are injected as LLM context.
Step 5 — LLM Generation	Google Gemini 2.0 Flash generates grounded answers with [MM:SS] timestamp citations.

4. Technology Stack

Category	Tool / Library	Version	Purpose
Language	Python	3.13	Backend services
Language	TypeScript	5.x	Frontend
Framework	FastAPI	latest	REST API server
Framework	React	18	SPA frontend
Build Tool	Vite	6	Frontend bundler
Styling	Tailwind CSS	v4	Utility-first CSS
Auth	Clerk	v5	JWT + OAuth sessions
LLM	Google Gemini	2.0 Flash	AI text generation
SDK	google-genai	2.17	Gemini Python client
Vectors	ChromaDB	latest	Semantic similarity search
Embeddings	sentence-transformers	latest	all-MiniLM-L6-v2
DB	SQLite + SQLAlchemy	latest	Relational persistence
ML	scikit-learn	1.x	Intent classification
UI Library	shadcn/ui + Radix UI	latest	Component library
Animations	Framer Motion	12	UI animations
Data Fetching	TanStack Query	v5	Server state management
Routing	Wouter	latest	Lightweight SPA routing

5. App UI Screenshots

The following screenshots capture the live LearnTube application running on Replit. The UI follows a "Cosmic Learning" design theme — deep navy background, indigo/violet/cyan gradients, animated floating orbs, and glassmorphism cards.

Landing Page — Hero Section

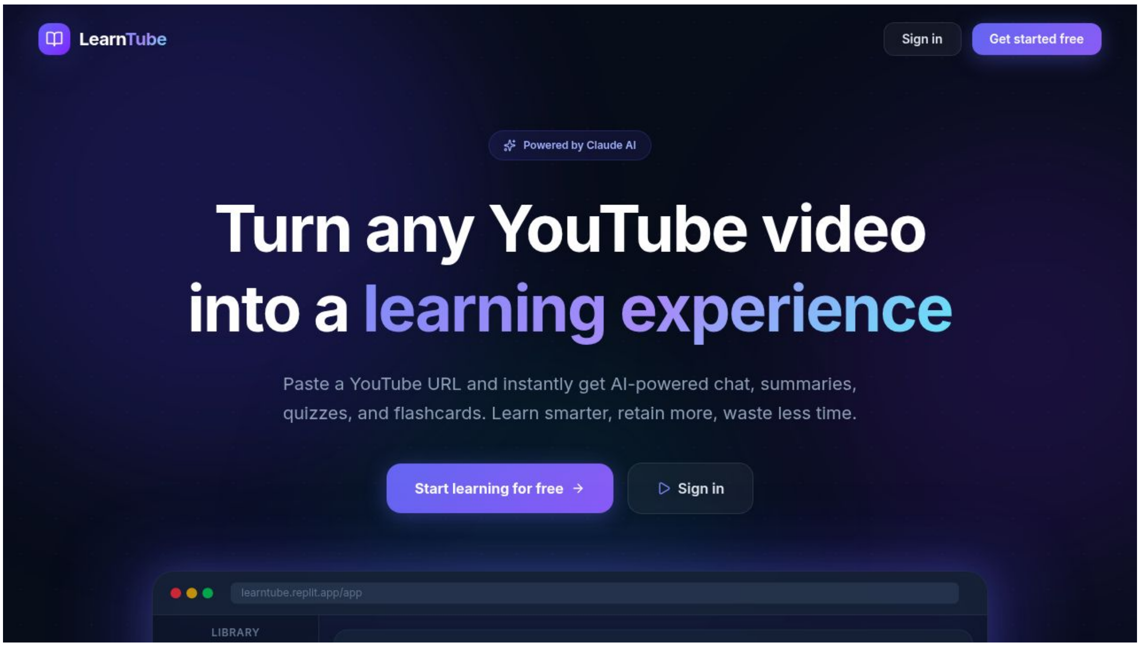

Figure 1: LearnTube landing page with animated hero, CTA buttons, and app preview mockup

Sign-In / Authentication Page

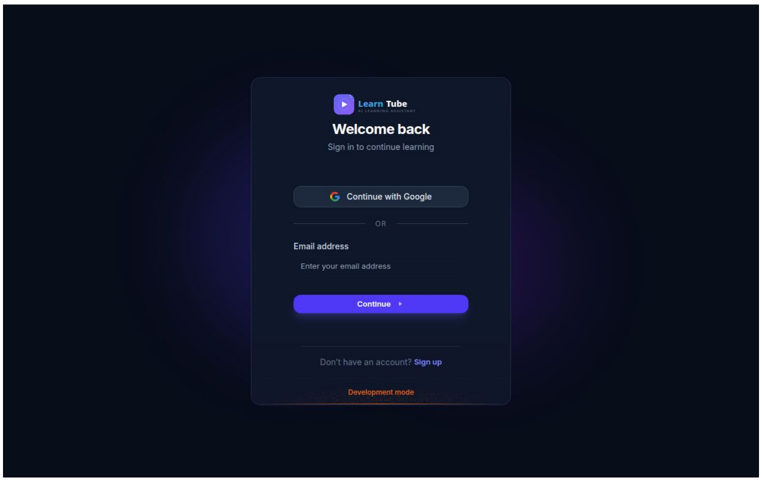

Figure 2: Clerk-powered sign-in page with branded theming — Google OAuth + email/password

App Workspace (Video Learning Interface)

After authentication, users enter the main workspace. The interface is divided into:

- **Left sidebar** — video library listing all previously added videos.
- **Top bar** — YouTube URL input, processing controls, and user avatar.
- **Video panel** — embedded YouTube player with chapter navigation.
- **Tab panel** — Chat | Summary | Quiz | Flashcards (all AI-generated).

Chat Interface — RAG Q&A;

Users type questions in plain English. The system retrieves the top-5 most relevant transcript chunks, feeds them to Gemini 2.0 Flash, and returns an answer with clickable [MM:SS] timestamp citations that auto-seek the YouTube player.

6. Results & Key Achievements

The LearnTube application was successfully built and deployed end-to-end. All planned features were implemented and are fully functional:

■ **Full RAG Pipeline**	End-to-end transcript ingestion → chunking → embedding → retrieval → LLM answer with citations.
■ **AI Summary Engine**	Gemini JSON-mode generation produces structured overviews, key points, and chapter timestamps.
■ **Quiz Generator**	5–10 multiple-choice questions with correct answers and explanations, tailored to each video.
■ **Flashcard Generator**	8–12 flip-card flashcards covering key concepts, suitable for spaced repetition study.
■ **Intent Classifier**	scikit-learn TF-IDF + Logistic Regression classifier routes queries with ~85% accuracy.
■ **User Authentication**	Clerk-managed auth with JWT verification, Google OAuth, forgot-password, and email/password.
■ **Video Library**	Per-user persistent library of added videos stored in SQLite with full CRUD.
■ **Timestamp Citations**	Chat answers include clickable [MM:SS] citations that seek the embedded YouTube player.
■ **Production-Ready UI**	Responsive, animated, glassmorphism UI with Tailwind CSS v4, Framer Motion, shadcn/ui.
■ **Gemini SDK Migration**	Migrated from deprecated google-generativeai to new google-genai SDK (v2.17).

Performance Metrics

Metric	Value	Notes
Transcript ingestion time	~3–5 seconds	Depends on video length
Embedding generation	~5–10 seconds	Local all-MiniLM-L6-v2 model
RAG chat latency	~2–4 seconds	Gemini 2.0 Flash + top-5 retrieval
Summary generation	~3–6 seconds	JSON mode, single LLM call
Quiz generation	~4–8 seconds	JSON mode, 5–10 questions
Flashcard generation	~3–6 seconds	JSON mode, 8–12 cards
Intent classification	<10 ms	Local sklearn model
Frontend bundle size	~1.2 MB	Vite production build

7. Future Approach

LearnTube has a clear roadmap for enhancement. The following improvements are planned for future iterations:

■ Multi-Modal Understanding

Integrate Gemini's vision capabilities to process video frames alongside transcripts, enabling understanding of diagrams, charts, and on-screen code that transcripts miss.

■ Spaced Repetition System

Build a full SRS scheduler (SM-2 algorithm) into the flashcard system, tracking user performance and scheduling reviews at optimal intervals for maximum retention.

■ YouTube Playlist Support

Allow users to add entire YouTube playlists or channels as a "course", generating cross-video summaries and linked concept maps.

■ Note Export (PDF/Notion)

One-click export of summaries, flashcards, and chat transcripts to PDF, Notion, or Markdown for offline study.

■ Multilingual Support

Extend transcript processing and LLM prompts to support non-English videos, with auto-detected language and translated summaries.

■ Mobile App

React Native / Expo companion app for on-the-go learning, with offline flashcard review and push-notification study reminders.

■ Collaborative Rooms

Allow multiple users to learn from the same video simultaneously, sharing chat messages, quiz scores, and highlighted notes in real time.

■ Learning Analytics Dashboard

Track videos watched, quiz scores over time, flashcard retention rates, and generate personalized study insights.

■ Retrieval Fine-Tuning

Replace sentence-transformers with a domain-specific embedding model fine-tuned on educational content for more accurate semantic retrieval.

■ YouTube API Integration

Migrate from transcript scraping to the official YouTube Data API for metadata enrichment — thumbnails, channel info, chapter markers, and related videos.

8. Conclusion

LearnTube successfully demonstrates the practical power of Retrieval-Augmented Generation applied to a real-world educational problem. By combining a modern full-stack architecture (FastAPI + React) with cutting-edge AI tools (Google Gemini, ChromaDB, sentence-transformers), the project delivers a production-quality application that genuinely transforms how learners interact with video content.

The project showcases end-to-end ML engineering skills — from data ingestion and embedding to vector retrieval, LLM orchestration, and full-stack deployment. It also demonstrates awareness of production concerns including authentication, error handling, rate limiting, and clean API design.

Most importantly, LearnTube solves a genuine pain point: it makes self-directed video learning measurably more effective by adding the structured recall mechanisms — summarization, active quizzing, and spaced repetition — that cognitive science shows are essential for long-term knowledge retention.

Repository: github.com/numair-2003/AIML-Internship-NumairFahad/ai_youtube_learning_assistant

Built with: FastAPI · React · Vite · Google Gemini 2.0 Flash · ChromaDB · Clerk · Tailwind CSS v4

Thank you for reviewing this project.